

ACR43989

## 2-Pyridineboronic acid N-phenyldiethanolamine ester

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 吡啶-2-硼酸 N-苯胺乙醇酯  
2-Pyridineboronic acid N-phenyldiethanolamine ester

**Cat No. :** 439890000; 439890010; 439890050; 439890100  
**Molecular Formula** C15 H17 B N2 O2

**Supplier**  
**UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG, United Kingdom

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Solid

**Appearance**  
White

**Odor**  
No information available

#### Emergency Overview

May cause drowsiness and dizziness. Causes skin irritation. Causes serious eye damage. May cause respiratory irritation. May cause damage to organs through prolonged or repeated exposure.

#### Classification of the substance or mixture

|                                                      |            |
|------------------------------------------------------|------------|
| Flammable liquids.                                   | Category 3 |
| Skin Corrosion/Irritation                            | Category 2 |
| Serious Eye Damage/Eye Irritation                    | Category 1 |
| Specific target organ toxicity - (single exposure)   | Category 3 |
| Specific target organ toxicity - (repeated exposure) | Category 2 |

#### Label Elements



# SAFETY DATA SHEET

## 2-Pyridineboronic acid N-phenyldiethanolamine ester

**Signal Word****Danger****Hazard Statements**

H226 - Flammable liquid and vapor  
H336 - May cause drowsiness or dizziness  
H315 - Causes skin irritation  
H318 - Causes serious eye damage  
H335 - May cause respiratory irritation  
H373 - May cause damage to organs through prolonged or repeated exposure

**Precautionary Statements****Prevention**

P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

Vapors may cause flash fire or explosion.

**Health Hazards**

May cause drowsiness or dizziness. Causes skin irritation. May cause respiratory irritation. May cause damage to organs through prolonged or repeated exposure.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. .

**Other Hazards**

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component                                           | CAS No      | Weight % |
|-----------------------------------------------------|-------------|----------|
| 2-Pyridineboronic acid N-phenyldiethanolamine ester | 662138-96-7 | 50-70    |
| Isopropyl alcohol                                   | 67-63-0     | 15-30    |
| N-Phenyl diethanolamine                             | 120-07-0    | 10-25    |

**SECTION 4. FIRST AID MEASURES****General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

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**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Most important symptoms and effects**

Causes eye burns. Causes severe eye damage.

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

### SECTION 5. FIRE-FIGHTING MEASURES

**Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. Flammable.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### SECTION 6. ACCIDENTAL RELEASE MEASURES

**Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

**Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

### SECTION 7. HANDLING AND STORAGE

**Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place. Flammables area.

**Specific Use(s)**

Use in laboratories

### SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

## 2-Pyridineboronic acid N-phenyldiethanolamine ester

## Control Parameters

| Component         | China                                                     | Taiwan                                     | Thailand     | Hong Kong                                                                                   |
|-------------------|-----------------------------------------------------------|--------------------------------------------|--------------|---------------------------------------------------------------------------------------------|
| Isopropyl alcohol | TWA: 350 mg/m <sup>3</sup><br>STEL: 700 mg/m <sup>3</sup> | TWA: 400 ppm<br>TWA: 983 mg/m <sup>3</sup> | TWA: 400 ppm | TWA: 400 ppm<br>TWA: 983 mg/m <sup>3</sup><br>STEL: 500 ppm<br>STEL: 1230 mg/m <sup>3</sup> |

| Component         | ACGIH TLV                     | OSHA PEL                                                                                                                                                                          | NIOSH                                                                                                         | The United Kingdom                                                                                                  | European Union |
|-------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------|
| Isopropyl alcohol | TWA: 200 ppm<br>STEL: 400 ppm | (Vacated) TWA: 400 ppm<br>(Vacated) TWA: 980 mg/m <sup>3</sup><br>(Vacated) STEL: 500 ppm<br>(Vacated) STEL: 1225 mg/m <sup>3</sup><br>TWA: 400 ppm<br>TWA: 980 mg/m <sup>3</sup> | IDLH: 2000 ppm<br>TWA: 400 ppm<br>TWA: 980 mg/m <sup>3</sup><br>STEL: 500 ppm<br>STEL: 1225 mg/m <sup>3</sup> | STEL: 500 ppm 15 min<br>STEL: 1250 mg/m <sup>3</sup> 15 min<br>TWA: 400 ppm 8 hr<br>TWA: 999 mg/m <sup>3</sup> 8 hr |                |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

## Exposure Controls

## Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

## Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material                                      | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|-----------------------------------------------------|-----------------------------------|-----------------|-------------|-----------------------|
| Nitrile rubber<br>Neoprene<br>Natural rubber<br>PVC | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** No protective equipment is needed under normal use conditions.

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

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Small scale/Laboratory use      Maintain adequate ventilation

**Hygiene Measures**      Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**      No information available.

### SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>                              | White                    |                                          |
| <b>Physical State</b>                          | Solid                    |                                          |
| <b>Odor</b>                                    | No information available |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>pH</b>                                      | No information available |                                          |
| <b>Melting Point/Range</b>                     | > 300 °C / > 572 °F      |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | Not applicable           |                                          |
| <b>Flash Point</b>                             | No information available | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | Not applicable           | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available |                                          |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Vapor Pressure</b>                          | No data available        |                                          |
| <b>Vapor Density</b>                           | Not applicable           | Solid                                    |
| <b>Specific Gravity / Density</b>              | No data available        |                                          |
| <b>Bulk Density</b>                            | No data available        |                                          |
| <b>Water Solubility</b>                        | No information available |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Component</b>                               | <b>log Pow</b>           |                                          |
| Isopropyl alcohol                              | 0.05                     |                                          |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>Viscosity</b>                               | Not applicable           | Solid                                    |
| <b>Explosive Properties</b>                    | No information available |                                          |
| <b>Oxidizing Properties</b>                    | No information available |                                          |
| <b>Molecular Formula</b>                       | C15 H17 B N2 O2          |                                          |
| <b>Molecular Weight</b>                        | 268.13                   |                                          |

### SECTION 10. STABILITY AND REACTIVITY

|                                 |                                                                               |
|---------------------------------|-------------------------------------------------------------------------------|
| <b>Stability</b>                | Stable under recommended storage conditions.                                  |
| <b>Hazardous Reactions</b>      | None under normal processing.                                                 |
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur.                                      |
| <b>Conditions to Avoid</b>      | Incompatible products. Excess heat. Avoid dust formation. Protect from light. |
| <b>Materials to avoid</b>       | Strong oxidizing agents. Acids.                                               |

**Hazardous Decomposition Products** Thermal decomposition can lead to release of irritating gases and vapors. Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>). Oxides of boron.

### SECTION 11. TOXICOLOGICAL INFORMATION

# SAFETY DATA SHEET

## 2-Pyridineboronic acid N-phenyldiethanolamine ester

### Product Information

#### (a) acute toxicity; Toxicology data for the components

| Component               | LD50 Oral                                  | LD50 Dermal         | LC50 Inhalation             |
|-------------------------|--------------------------------------------|---------------------|-----------------------------|
| Isopropyl alcohol       | 5045 mg/kg ( Rat )<br>3600 mg/kg ( Mouse ) | 12800 mg/kg ( Rat ) | 72.6 mg/L ( Rat ) 4 h       |
| N-Phenyl diethanolamine | LD50 = 980 mg/kg ( Rat )                   |                     | LC50 > 0.1 mg/L ( Rat ) 8 h |

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 1

#### (d) respiratory or skin sensitization;

Respiratory No data available  
Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

(g) reproductive toxicity; No data available

(h) STOT-single exposure; Category 3

Results / Target organs  
Respiratory system  
Central nervous system (CNS)

(i) STOT-repeated exposure; Category 2

Target Organs Blood.

(j) aspiration hazard; Not applicable  
Solid

Symptoms / effects, both acute and delayed No information available

## SECTION 12. ECOLOGICAL INFORMATION

### Ecotoxicity effects

| Component         | Freshwater Fish                                                                                                                                                      | Water Flea                                      | Freshwater Algae                                                                                     | Microtox                                           |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Isopropyl alcohol | LC50: = 9640 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: > 1400000 µg/L, 96h (Lepomis macrochirus)<br>LC50: = 11130 mg/L, 96h static (Pimephales promelas) | 13299 mg/L EC50 = 48 h<br>9714 mg/L EC50 = 24 h | EC50: > 1000 mg/L, 72h (Desmodesmus subspicatus)<br>EC50: > 1000 mg/L, 96h (Desmodesmus subspicatus) | = 35390 mg/L EC50 Photobacterium phosphoreum 5 min |

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## 2-Pyridineboronic acid N-phenyldiethanolamine ester

|                         |                                                                |  |  |  |
|-------------------------|----------------------------------------------------------------|--|--|--|
|                         | LC50: = 10000000 µg/L,<br>96h (Daphnia)                        |  |  |  |
| N-Phenyl diethanolamine | LC50: = 735 mg/L, 96h<br>flow-through<br>(Pimephales promelas) |  |  |  |

**Persistence and Degradability** No information available

**Bioaccumulative Potential** No information available

| Component         | log Pow | Bioconcentration factor (BCF) |
|-------------------|---------|-------------------------------|
| Isopropyl alcohol | 0.05    | No data available             |

**Mobility in soil** No information available

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 13. DISPOSAL CONSIDERATIONS

**Waste from Residues/Unused Products** Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains.

## SECTION 14. TRANSPORT INFORMATION

Road and Rail Transport

**UN-No** UN3175  
**Proper Shipping Name** Solids containing flammable liquid, n.o.s  
**Technical Shipping Name** Isopropyl alcohol  
**Hazard Class** 4.1  
**Packing Group** II

IMDG/IMO

**UN-No** UN3175  
**Proper Shipping Name** Solids containing flammable liquid, n.o.s  
**Technical Shipping Name** Isopropyl alcohol  
**Hazard Class** 4.1  
**Subsidiary Hazard Class** +  
**Packing Group** II

IATA

**UN-No** UN3175  
**Proper Shipping Name** Solids containing flammable liquid, n.o.s  
**Technical Shipping Name** Isopropyl alcohol

## 2-Pyridineboronic acid N-phenyldiethanolamine ester

**Hazard Class** 4.1  
**Packing Group** II

**Special Precautions for User** No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component               | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Isopropyl alcohol       | X                                                   | X                                       | X    | X     | 200-661-7 | X    | X   | X     | X    | X    | X    | KE-29363 |
| N-Phenyl diethanolamine | -                                                   | -                                       | X    | X     | 204-368-5 | X    | X   | X     | X    | X    | X    | KE-28387 |

## National Regulations

## SECTION 16. OTHER INFORMATION

**Creation Date** 21-Jan-2011  
**Revision Date** 12-Apr-2024  
**Revision Summary** Not applicable.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships



## 2-Pyridineboronic acid N-phenyldiethanolamine ester

OECD - Organisation for Economic Co-operation and Development  
BCF - Bioconcentration factor

ATE - Acute Toxicity Estimate  
VOC - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Physical hazards**

On basis of test data

**Health Hazards**

Calculation method

**Environmental hazards**

Calculation method

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**